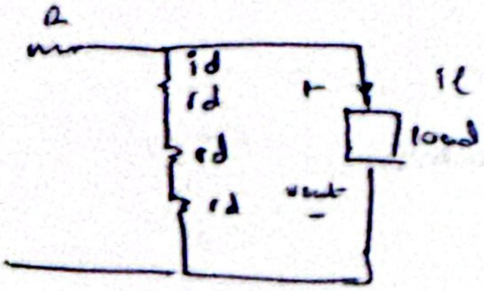
(c1)

circuit can be simplified as:



$r_d = \frac{V_T}{I_D} = \frac{26\,mV}{5\,mA} = 5.2\,\Omega$ from eq 3.60

since $i_L = +1\,mA$

$i_d = -1\,mA$

change in Vout: $V_{out} = (-1\,mA)(3 \times 5.2)$

$= -15.6\,mV$

(Q2)

$$V_R \simeq \frac{1}{2} \frac{V_P - 5V_{DON}}{R_L C_1 f_{in}}$$

$V_P = 5$ volts

$R_L = 1k\Omega$

$C_1 = 100\mu F$ Plug in the values.

$f_{in} = 50$ Hz

$$\frac{1}{2} \frac{5 - 5(0.8)}{1000 \times 100\times10^{-6} \times 50}$$

= ~~150mV or 0.15volts~~. 100 mV or 0.1volts.

(b) The bias current through the diodes is the same as the bias current through R_1.

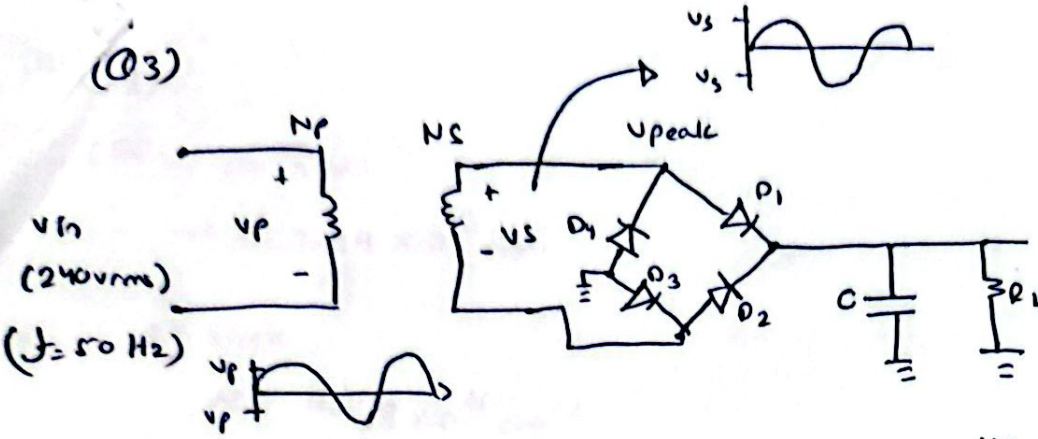
$$\frac{V_P - 5(0.8)}{R_1} = 1mA$$

$$r_d = \frac{V_T}{I_D} = 26\Omega$$

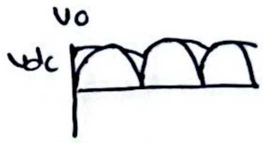
$$V_{Rload} = \frac{3r_d}{R_1 + 3r_d} V_R = \cancel{V_P = 126.06mV}$$

$V_{R\,load} = 7.23 \times 10^{-3}$

(Q3)



Note: 150 Ω used
Same calculation but with 100 Ω
Vpeak should be done



$V_{out} = V_{dc} = 15V$

$R_L = 150\Omega$

ripple = 1 %

$I_{dc} = \frac{V_{dc}}{R_L} = \frac{15}{150} = 0.1A$

Idiode:

Avg peak $= V_{dc}/150\Omega = 100mA$

$= V_{peak}/150\Omega = 157.1mA$

capacitor:

$V_R \geq 1\%$ of $V_{dc} = 0.15V$

$V_R = \frac{I_L}{2cf_{in}} \Rightarrow C = \frac{I_L}{2V_R f_{in}}$

$C \geq 6.667mF$

we know that $V_{dc} = \frac{2}{\pi} V_{peak}$ (or $I_{dc} = \frac{2}{\pi} I_{peak}$)

$I_{peak} = \frac{\pi}{2} I_{dc} = \frac{\pi}{2}(0.1) = 0.1571A$

$V_{peak} = \frac{\pi}{2} V_{dc} = \frac{\pi}{2}(15) = 23.565V$

two diodes in the complete path results in voltage drop across two diodes i.e, $V_{diode\ drop} = 2 \times 0.7 = 1.4V$

$V_{peak} = V_{peak} + V_{diode\ drop} = 23.565 + 1.4 =$

$V_{peak} = 24.965V = V_s$ (voltage at secondary transformer)

$V_p = 240\ V_{rms}$

$V_p = \sqrt{2} \times 240$ (peak)

$V_p = 339.411V$

Turn ratio

$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{339.411}{24.965} = N_p : N_s = 13.6 : 1$ (step down)

PIV rating = 1.5 to 2 times Vpeak =

$2 \times 24.965 = 49.93 \approx 50V$

$I_{d(max)}$ rating = 1.5 to 2 times Ipeak

$2 \times \frac{24.965}{150} = 0.1664 \approx 0.1667 A$

Q4)

(a) At 300k

$n_i = 2.49 \times 10^{13}\ cm^{-3}$

At 600k

$n_i = 4.149 \times 10^{16}\ cm^{-3}$

<u>for Si</u>

n_i at 300k $= 1.078 \times 10^{10}\ cm^{-3}$

n_i at 600k $= 1.526 \times 10^{15}\ cm^{-3}$

The intrinsic carriers are more in Ge than Si for both temperatures 300k and 600k.

(b) $n = N_D = 5 \times 10^{16}\ cm^{-3}$

at $T = 300k$

$n_i^2 = np$

$p = \frac{n_i^2}{N_D} = 1.25 \times 10^{10}\ cm^{-3}$

at $T = 600k$

$n_i^2 = np$

$p = \frac{n_i^2}{N_D} = 3.442 \times 10^{16}\ cm^{-3}$

(Q5)

a) Mobility in silicon electrons $= 1350\ cm^2/V\cdot s$

Mobility of holes in Si $= 480\ cm^2/V\cdot s$

$E_{field} = 0.1 \frac{V}{\mu m}$

velocity of electrons $= \mu_n E = \left(1350 \frac{cm^2}{V\cdot s}\right)\left(0.1 \frac{V}{\mu m}\right)$

$= 1.35 \times 10^{4}\ m/s$

velocity of holes $= \mu_p E = \left(480 \frac{cm^2}{V\cdot s}\right) \cdot \left(0.1 \frac{V}{\mu m}\right)$

$= 4.8 \cdot 10^3\ m/s$

(b)

$E = 0.1\ V/\mu m$ hole current is negligible as stated

$\mu_n = 1350\ cm^2/Vs$ $\mu_p = 480\ cm^2/Vs$

$J_{tot} = 1\ mA/\mu m^2 = q[\mu_n n E + \overset{\text{negligible}}{0}] = q\mu_n n E$

$n = \frac{J_{tot}}{q\mu_n n E} = \frac{1\ mA/\mu m^2}{(1.6\times10^{-19})(1350\ cm^2/Vs)(0.1\ V/\mu m)}$

$= 4.6 \times 10^{17}\ cm^{-3}$

Q6)

$L = 0.1\mu m \quad A = (0.05\mu m)^2 \quad V = 1V$

$\mu_n = 1350\ cm^2/Vs \quad \mu_p = 480\ cm^2/Vs$

$n = 10^{17} cm^{-3}$

(a)

$n_i(T=300k) = 5.2\times10^{15}(300)^{3/2} \exp\left[\frac{-1.12eV}{2(1.38\times10^{-23})\times(300)}\right]$

$= 1.077\times10^{10}$

$= 1.08\times10^{10}\ cm^{-3}$

$p = \frac{n_i^2}{n} = 1.17\times10^{3}\ cm^{-3} \quad E = V/L = 10\ V/\mu m$

$I_{tot} = A\cdot J_{tot} = A\cdot q[\mu_n n + \mu_p p]E$

$= A\cdot q[\mu_n n + \mu_p(n_i^2/n)]E$

$10(V/\mu m)\times(0.05\mu m)^2(1.6\times10^{-19})\left[1350\frac{cm^2}{Vs}(10^{17}cm^{-3}) + 480\frac{cm^2}{Vs}(1.17\times10^{3}cm^{-3})\right]$

$\cancel{\times 10(V/\mu m)}$

$I_{tot} = 0.054\ mA$

b) $400k$, $n_i = 3.7\times10^{12}cm^{-3}$, $p = n_i^2/n = 1.4\times10^{8}cm^{-3}$, $E = 10V/\mu m$

$I_{tot} = A\cdot q[\mu_n n + \mu_p(n_i^2/n)]E$

$(10\ V/\mu m)\times(0.05\mu m)^2\times1.6\times10^{-19}[1350\frac{cm^2}{Vs}(10^{17}cm^{-3}) + 480\frac{cm^2}{Vs}(1.4\times10^{8}cm^{-3}$

$I_{tot} = 0.054mA.$

(Q7)

$N_D = 3\times10^{16}$ $n_i = 1.08\times10^{10}\,cm^{-3}$

$V_0 = ?$

$$V_0 = \frac{kT}{q}\ln\left(\frac{N_D N_A}{n_i^2}\right) = \frac{kT}{q}\ln\left(\frac{N_D}{N_i}\right)$$

$$= \frac{(1.38\times10^{-23}\,J/K)(300)}{1.6\times10^{-19}}\ln\left(\frac{3\times10^{16}\,cm^{-3}}{1.08\times10^{10}\,cm^{-3}}\right)$$

$= 0.384$ volts.

(Q8)

$$\frac{C_{j0}}{\sqrt{1+\frac{0.5}{V_0}}} = 2.2 \rightarrow eq\ 1 \qquad \frac{C_{j0}}{\sqrt{1+\frac{1.5}{V_0}}} \rightarrow eq\ 2$$

$$1 \div 2 = \frac{1+1.5/V_0}{1+0.5/V_0} = \left(\frac{2.2}{1.3}\right)^2 = V_0 = 0.0365\,V$$

↳ put in eq 1

$$C_{j0} = 2.2\times\sqrt{1+\frac{0.5}{V_0}} = 8.43\ fF/\mu m^2$$

$$\frac{N_A N_D}{N_A+N_D} = (C_{j0})^2 \cdot V_0 \cdot \frac{2}{\varepsilon_{Si}\,q}$$

$$= \left(8.43\frac{fF}{\mu m^2}\right)^2 \times (0.0365\,V)\,\frac{2}{\varepsilon_{Si}\,q} \approx 3.13\times10^{17}\,cm^{-3}$$

$N_A > \frac{N_A N_D}{N_A+N_D} \triangleq y$ [fixing a value for N_A]

$$N_A = 2\cdot10^{18}\,cm^{-3} \Rightarrow N_D = \frac{y N_A}{N_A - y} = \frac{(3.13\times10^{17}\,cm^{-3})(2\cdot10^{18}\,cm^{-3})}{(2\cdot10^{18} - 3.13\times10^{17})\,cm^{-3}}$$

$$\approx 3.71\times10^{17}\,cm^{-3}$$

9)

(a) In forward bias, $I_D = 1mA$, $V_D = 750mV$

$\therefore I_S \simeq I_D e^{-V_D/V_T} = (1mA)(\exp)[-750mV/26mV]$

$= 2.97 \times 10^{-16} A$

b) Since $I_S \propto$ Area, doubling area implies doubling I_S

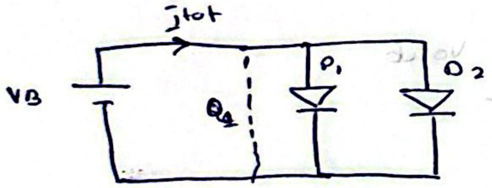
from a

$I_D = 1mA = 2 \times I_S e^{V_D/V_T}$

$V_D = V_T \ln\left(\frac{I_D}{2 I_S}\right) = (26mV) \ln\left(\frac{1mA}{2.297 \times 10^{-16} A}\right)$

$= 0.732$ volts

(b)



$I_{tot} = I_{D1} + I_{D2} = I_{S1}(e^{V_B/V_T} - 1) + I_{S2}(e^{V_B/V_T} - 1)$

$I_{tot} = (I_{S1} + I_{S2})(e^{V_B/V_T} - 1)$

Therefore the parallel combination operates as an exponential device, with an equivalent saturation current $= I_{S1} + I_{S2}$

By KVL $V_{D1} = V_{D2}$

$V_T \ln\left(\frac{I_{D1}}{I_{S1}}\right) = V_T \ln\left(\frac{I_{D2}}{V_{D2}}\right)$

Also, $I_{tot} = I_{D1} + I_{D2} \Rightarrow I_{D2} = I_{tot} - I_{D1}$

$$\therefore \quad V_T \ln\left(\frac{I_{D1}}{I_{S1}}\right) = V_T \ln\left(\frac{I_{tot} - I_{D1}}{I_{S2}}\right)$$

$$I_{D1} = I_{tot}\left(\frac{I_{S1}}{I_{S1} + I_{S2}}\right)$$

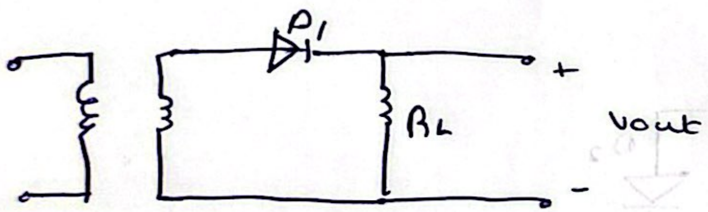
$$I_{D2} = I_{tot}\left(\frac{I_{S2}}{I_{S1} + I_{S2}}\right)$$

(Q10)

For a half wave rectifier:

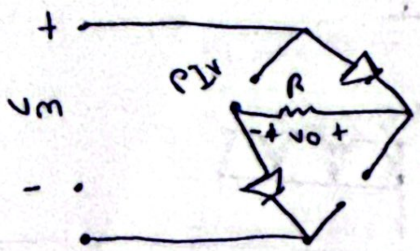
→ it allows only one half cycle of AC input to pass. When the negative cycle of AC input happens, the diode is reversed biased and the entire AC voltage (peak voltage) appears across the diode.

$$PIV = V_p.$$



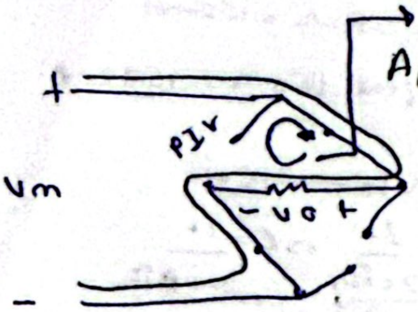
(Q10)(b)

PIV (full wave bridge rectifier).



diodes will be treated as short circuit

replace diodes with •——•



→ second loop

Apply kirchoff's law

$+vm - vo = 0$

$vm = vo$

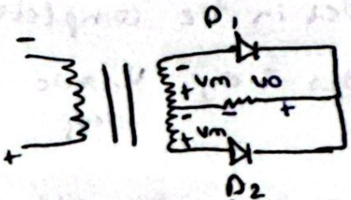
Apply kirchoff's law denoted by the arrow.

$-vo + PIV = 0$

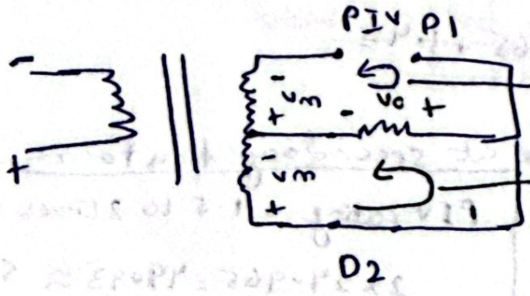
$-vm + PIV = 0$

$PIV = vm$ hence proved.

PIV (full wave center tapped circuit).



vo = voltage across resisto..



→ Apply kirchoff in this loop next

→ Apply kirchoff's in this loop.

$+vm - vo = 0$

$vm = vo$

$+vo - PIV + vm = 0$

$v\cancel{o}m - PIV + vm = 0$

$2vm = 2PIV$